

AI Emergency Patient Intake

Team Workflow and Role Ownership

A shared architecture proposal before implementation begins

Product goal

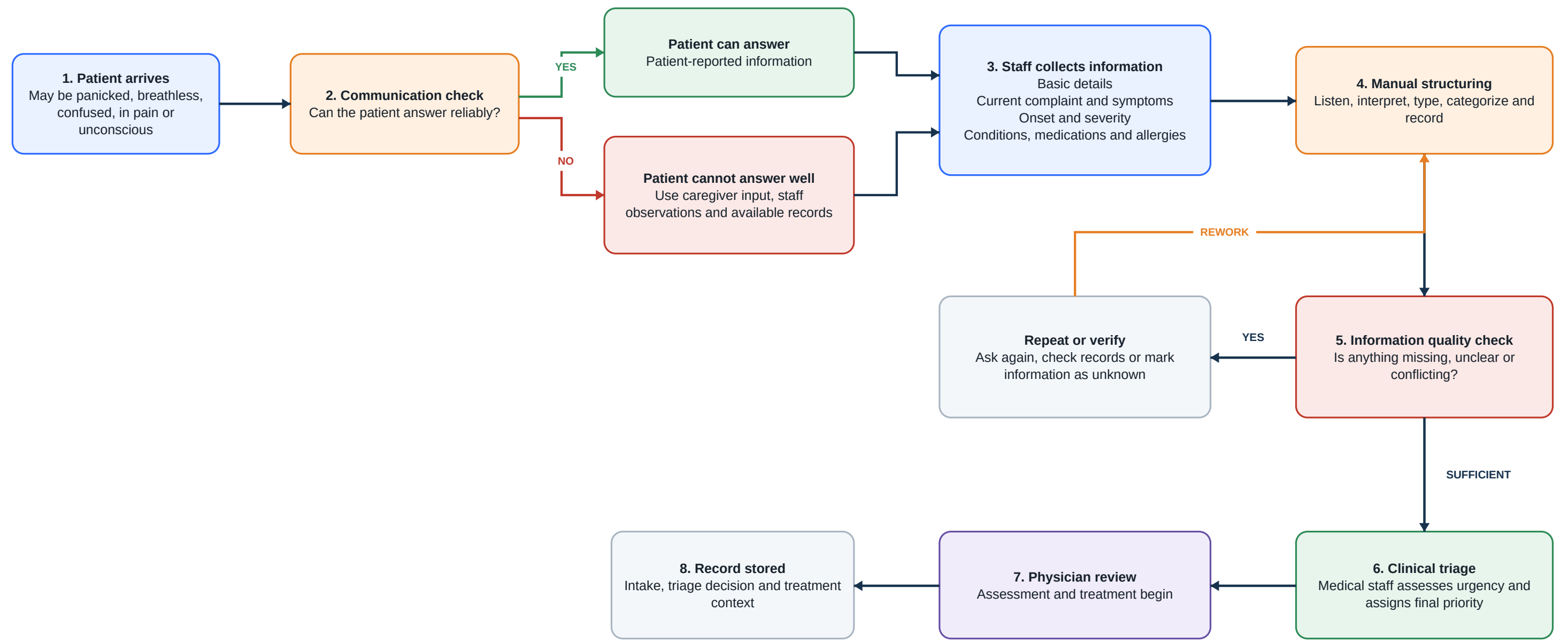
Reduce the information bottleneck between a distressed patient and emergency medical staff by converting natural conversation, caregiver input, staff observations and verified history into structured clinician-ready information.



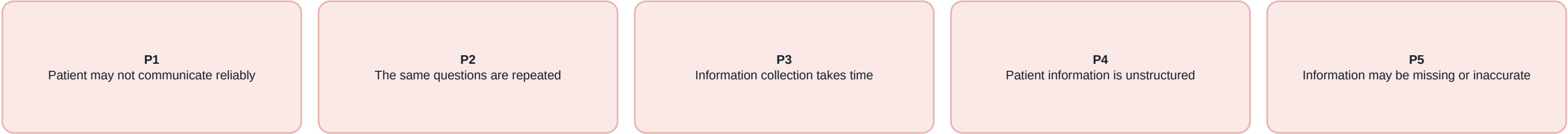
Decision gate: the team reviews and agrees on this structure before role-level implementation starts.

1. Current Emergency Intake Flow

Where the communication and documentation bottlenecks occur

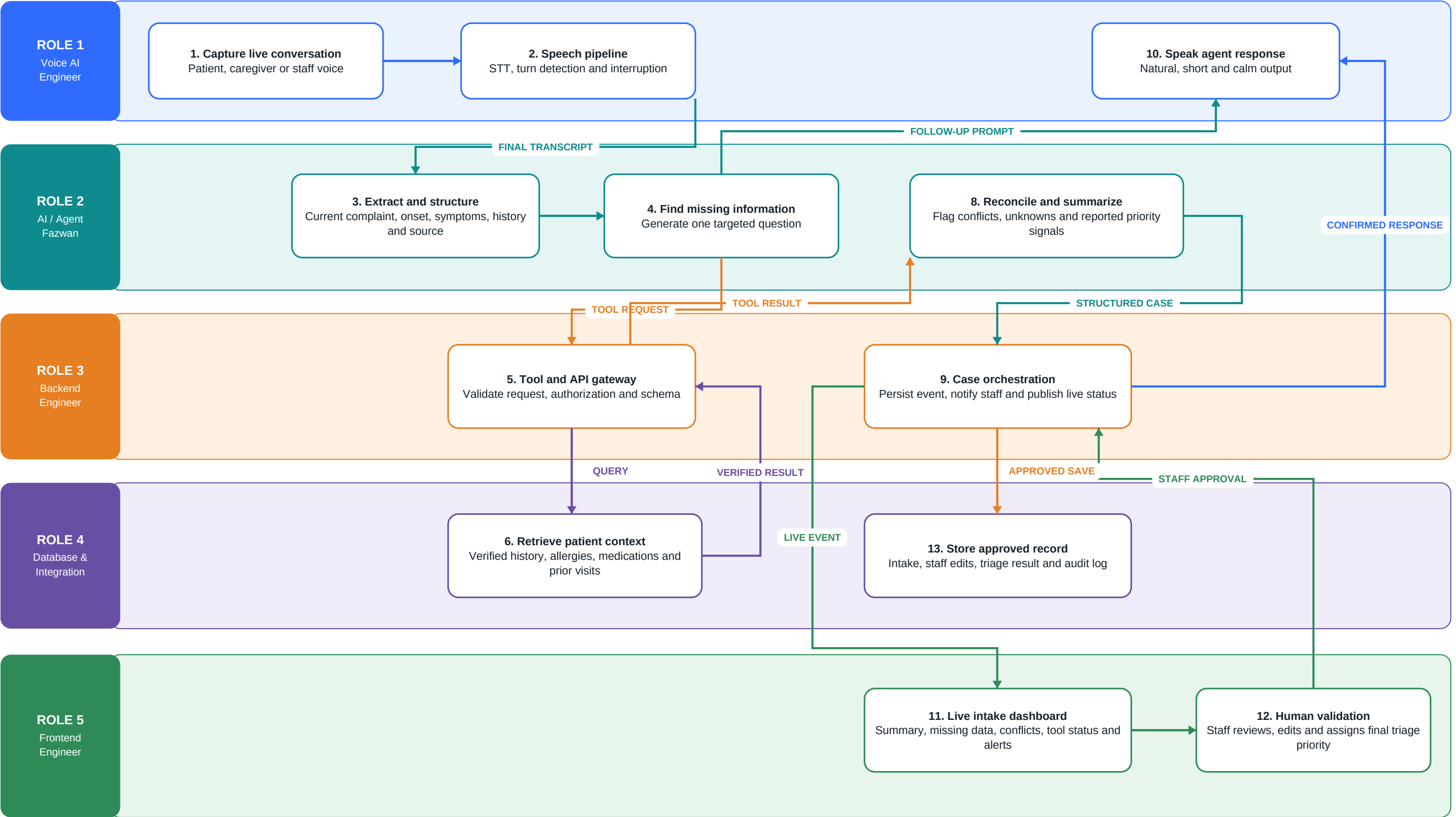


Pain points identified



2. Proposed AI-Assisted Workflow by Role

Each lane has one owner; arrows define the integration contracts



Human decision gate: AI structures and highlights reported information. Emergency medical staff remain responsible for clinical assessment and final triage priority.

3. Ownership and Handoff Contract

Use this page to prevent duplicate work and integration conflicts

Working rule: Every role owns its internal implementation, but all cross-role data must follow an agreed contract: event name, JSON schema, success response, error response and owner.

Role	Owns and delivers	Sends to other roles	Receives from other roles	Must not duplicate
Role 1 Voice AI Engineer	AssemblyAI connection; microphone/audio streaming; STT/TTS; turn detection; interruptions; session state.	Final transcript turns, session ID/status and spoken-response completion.	Agent response text, conversation state and approved prompt configuration from Role 2.	Triage logic, backend business logic, database schema or dashboard UI.
Role 2 AI / Agent Engineer Fazwan	System prompt; conversation flow; extraction schema; missing-information logic; source/confidence labels; tool policy; safety guardrails; structured summary.	Response instructions to Role 1; validated tool requests and structured case events to Role 3.	Final transcript from Role 1; verified tool result from Role 3; retrieved context through backend tools.	Direct database access, API implementation, UI code or the medical staff's final triage decision.
Role 3 Backend Engineer Askeri	FastAPI; authentication; endpoint contracts; tool execution; workflow orchestration; retries; staff notifications; event publishing.	Database queries/commands to Role 4; live case events to Role 5; confirmed tool results to Role 2.	Validated tool requests from Role 2; data responses from Role 4; staff actions from Role 5.	Agent prompt/decision policy, physical data model ownership or frontend rendering.
Role 4 Database & Integration Ishan	Patient schema; synthetic records; history and allergies; call logs; audit trail; permissions; privacy; retention; query performance.	Verified records, not-found responses, conflicts and save confirmations to Role 3.	Authorized read/write requests from Role 3 and approved staff updates routed by backend.	Calling the LLM directly, frontend UI, voice pipeline or backend workflow decisions.
Role 5 Frontend Engineer	Live monitoring dashboard; structured intake; source/confidence display; missing/conflict flags; staff edit, review and approval interface.	Staff corrections, approval, final triage selection and action requests to Role 3.	Live case events, tool status and structured intake data from Role 3.	Direct database writes, independent triage logic, voice pipeline or backend tool execution.

Source: Emergency Patient Intake - Problem Analysis & AI Opportunity Assessment. This is an architecture proposal for team agreement, not a clinical protocol.